

“ **THE ANIMAL NEVER COMES OUT ON SHORE... ITS SKIN IS BLACK AND THICK... ITS HEAD... IS SMALL, IT HAS NO TEETH, BUT ONLY TWO FLAT WHITE BONES.** ”

Georg Steller, German zoologist, 1740



Steller's sea cow was a large sea mammal that fed on kelp. Discovered in 1740 by German naturalist Georg Steller, it was extinct by 1767.

TOUGH AND RESISTANT TO CORROSION, STEEL is a practical metal for construction and machinery. But for thousands of years it was so hard to make reliably in any quantity that it was used only for blades. Then, in 1740, English clockmaker **Benjamin Huntsman** (1704–76) perfected his “**crucible**” method of making steel in Sheffield, England. It involved heating steel to 2912°F (1600°C) in a coke-fired furnace in clay pots or “crucibles” to make ingots of tough steel large and pure enough to cast into many shapes. Huntsman's crucible revolutionized steel making and over the next century, Sheffield's steel production rose from 200 to 80,000 tons per year. This was almost half of Europe's steel.

On June 4, 1740, Danish explorer **Vitus Bering** (1681–1741) launched an expedition to **map the remote Arctic coast of Siberia**. He sailed from Kamchatka in eastern Russia aboard the *St. Peter*, while fellow explorer **Aleksey Chirikov** (1703–48) sailed aboard the *St. Paul*. The ships became separated and **Bering discovered the Alaskan Peninsula** while **Chirikov found some of the Aleutian Islands**. After Bering fell ill with scurvy, his ship was wrecked on the Aleutians and he died there. Some of his crew built



ANDERS CELSIUS
(1701–41)

Born in Uppsala in Sweden, Anders Celsius succeeded his father as professor of astronomy at Uppsala University in 1730. He is most famous for devising the temperature scale that now bears his name, but he also helped discover the link between magnetic storms in the Sun and the aurora phenomenon on Earth.

a small boat and returned to Russia with news of fur-trading possibilities that would make Russia rich. Among the survivors was **German naturalist Georg Steller** (1709–46), who had collected **specimens of hitherto unknown species** of wildlife during the expedition. Steller's sea cow, Steller's jay, Steller's sea eagle, and Steller's eider all bear

his name. Within 27 years of its discovery, Steller's sea cow had been hunted to extinction.

Developed in the early 17th century by innovators such as the physician **Robert Fludd** and astronomer **Galileo Galilei**, the **thermometer** displayed temperature by showing the level of liquid in a glass tube as it expanded or contracted. But a century on, it had still not been agreed how to calibrate it. Among other suggestions, English physicist and mathematician Isaac Newton proposed a scale with the melting point of snow at one end and the boiling point of water at the other, with points divided by 33 degrees in between. In the end, the winner turned out to be the **temperature scale invented by Swedish astronomer Anders Celsius** (1701–44) in 1742, which developed into the **modern Celsius scale** (known until 1948 as Centigrade). In Celsius's scale, the two end points were set 100 degrees apart, with 100 degrees signifying the freezing point of water, while 0 degrees signified boiling point at standard atmospheric pressure. Two years later, Swedish naturalist **Carl Linnaeus** adopted Celsius's scale for his greenhouse thermometers, but reversed the scale so that 100 degrees was the boiling point of water.

6 THE NUMBER OF SPECIES **GEORG STELLER** DISCOVERED DURING THE 1740 VOYAGE

In 1742, French mathematician **Jean Le Rond D'Alembert** (1717–83) found another way to consider Newton's Second Law of Motion, by introducing a fictitious balancing or “equilibrium” force. Known as **D'Alembert's principle**, this **made calculations about dynamic or changing**

forces simpler by reducing them to static calculations. Also this year, American inventor and statesman **Benjamin Franklin** designed a **cast-iron stove** that could be set in the middle of a room to **maximize its heating effect**. Cast-iron stoves soon became immensely popular.

TEMPERATURE SCALES

In the 1700s, many temperature scales were used, including Réaumur's. Now just three are common: the Kelvin (K, introduced in 1848), Celsius (C), and Fahrenheit (F) scales. Each just shows degrees between fixed points. The Kelvin scale starts at absolute zero. One Kelvin is equal to one degree Celsius, so 273.15 K is 0°C, the melting point of water, and 373.15 K is 100°C, the boiling point of water.

	KELVIN	CELSIUS	FAHRENHEIT
	373K	100°C	212°F
	300K	27°C	81°F
	273K	0°C	32°F
	255K	–18°C	0°F
	200K	–73°C	–99°F
	100K	–173°C	–279°F
Absolute zero	0K	–273°C	–460°F

1740 British clockmaker Benjamin Huntsman develops crucible steel making

May 1741 Danish explorer Vitus Bering maps the coast of Alaska and Arctic Siberia

July 1741 Russian explorer Aleksey Chirikov discovers the Aleutian Islands

July 20, 1741 German naturalist Georg Steller lands in Alaska

1742 Swedish astronomer Anders Celsius proposes a centigrade temperature scale